CONFIDENTIAL — CUSTOMER DELIVERABLE

Installation and Configuration Record

Post-Deployment As-Built Package — azlocal-iic-01

Mode: as-built | Ranger Version: 1.6.0 | Generated: 04/06/2026 12:08:32

Table of Contents

- Document Control
- Run Summary
- Readiness Snapshot
- Environment Overview
- Topology and Operating Model
- Domain Coverage
- Node Inventory
- Collector Status
- Operational Risk Summary
- Workload Summary
- Capacity Summary
- Priority Recommendations
- VM Inventory
- VM Density Metrics
- Storage Pool Capacity
- ESU Enrollment
- Coverage Assessment
- Security Posture Summary
- Management Tool Coverage
- Installation Register (Bill of Materials)
- Per-Node Configuration Record
- Network Address Allocation Record
- Storage Configuration Record
- Identity and Security Record
- Azure Integration Record
- Validation Record
- Known Issues and Deviations
- Acceptance and Sign-Off

Priority Recommendations

- **[WARNING]** One or more node certificates expire within 90 days
  Review certificate ownership and renew expiring node certificates before handoff.
- **[WARNING]** iDRAC firmware below recommended baseline on all nodes
  Update iDRAC firmware via Dell OME or Lifecycle Controller during next maintenance window.
- **[INFORMATIONAL]** Azure Policy assignments discovered at resource group scope
  Verify policy scope extends to arc resource group and monitor for compliance drift.
- **[INFORMATIONAL]** Cluster event history includes a transient network quorum warning
  Review switch port configuration to confirm management NICs are on dedicated VLANs and bonded correctly.

Collector Overview

hardware

Status: success

Targets:

management-performance

Status: success

Targets:

monitoring-observability

Status: success

Targets:

storage-networking

Status: success

Targets:

topology-cluster

Status: success

Targets:

waf-assessment

Status: success

Targets:

workload-identity-azure

Status: success

Targets:

Document Control

Document Title	Azure Local As-Built Documentation — azlocal-iic-01
Package ID	azlocal-iic-01-as-built-20260417T001248Z
Report Tier	management
Revision	1.0 (initial handoff)
Classification	CONFIDENTIAL — CUSTOMER DELIVERABLE
Prepared By Azure Local Ranger v1.6.0	
Prepared On	04/06/2026 12:08:32
Schema Version	1.1.0-draft
Document Status	FINAL — AS-BUILT HANDOFF

Run Summary

- Mode: as-built
- Generated: 04/06/2026 12:08:32
- Cluster: azlocal-iic-01
- Nodes: 3
- Collectors: 7/7 successful
- Artifacts currently recorded: 0

Readiness Snapshot

- Schema validation: passed
- Critical findings: 0
- Warning findings: 2
- Informational findings: 2
- Successful collectors: 7 of 7
- Partial or failed collectors: 0

Environment Overview

- VMs discovered: 5
- Azure resources discovered: 10
- Connected Azure auth method: service-principal
- Monitoring resources: 5
- Management services running: 4

Topology and Operating Model

- Deployment type: hyperconverged
- Identity mode: ad
- Connectivity mode: connected
- Storage architecture: storage-spaces-direct
- Network architecture: switched
- Variant markers: connected

Domain Coverage

- Cluster nodes: 3
- Cluster roles: 5
- Storage pools: 1
- Physical disks: 24
- Network adapters: 12
- VMs discovered: 5
- Azure resources: 10
- Alerting resources: 2
- Management services: 4

Node Inventory

Node	Model	State	OS	OS Build	CPU Sockets	RAM (GiB)
azl-iic-n01						
PowerEdge R760						
Up						
—						
—						
—						
512						
azl-iic-n02						
PowerEdge R760						
Up						
—						
—						
—						
512						
azl-iic-n03						
PowerEdge R760						
Up						
—						
—						
—						
512						

Collector Status

- hardware: success ()
- management-performance: success ()
- monitoring-observability: success ()
- storage-networking: success ()
- topology-cluster: success ()
- waf-assessment: success ()
- workload-identity-azure: success ()

Operational Risk Summary

- Nodes not healthy: 0
- Unhealthy disks: 0
- High CPU nodes: 0
- Certificates expiring within 90 days: 1
- Azure Policy assignments: 2
- Schema warnings: 0
- Azure Advisor recommendations: total, 3 high-impact

Workload Summary

- Total VMs: 5
- Total nodes: 3
- AKS clusters: 1
- Arc-connected machines: 1
- Update compliance: No update resources tracked
- Licensing: Not collected
- VMs using Arc IP fallback: Not collected

Capacity Summary

- Storage total raw: 0 TiB
- Storage total usable: 0 TiB
- Storage used by workloads: 0 TiB (0% of usable)
- Storage reserve target: 0 TiB (0% of usable)
- Storage free usable: 0 TiB
- Projected safe allocatable: 0 TiB
- Thin provisioning ratio: Not computed
- vCPU:pCPU overcommit ratio: Not computed
- Memory overcommit ratio: Not computed
- Average VMs per node: N/A

Priority Recommendations

- [WARNING] One or more node certificates expire within 90 days: Review certificate ownership and renew expiring node certificates before handoff.
- [WARNING] iDRAC firmware below recommended baseline on all nodes: Update iDRAC firmware via Dell OME or Lifecycle Controller during next maintenance window.
- [INFORMATIONAL] Azure Policy assignments discovered at resource group scope: Verify policy scope extends to arc resource group and monitor for compliance drift.
- [INFORMATIONAL] Cluster event history includes a transient network quorum warning: Review switch port configuration to confirm management NICs are on dedicated VLANs and bonded correctly.

VM Inventory

Name	State	vCPU	RAM (GiB)	Host Node	Generation
avd-iic-sh01					
Running					
8					
16					
azl-iic-n01					
2					
avd-iic-sh02					
Running					
8					
16					
azl-iic-n02					
2					
avd-iic-sh03					
Running					
8					
16					
azl-iic-n03					
2					
arc-iic-vm01					
Running					
4					
8					
azl-iic-n01					
2					
arc-iic-vm02					
Running					
4					
8					

Name	State	vCPU	RAM (GiB)	Host Node	Generation
azl-iic-n02					
2					

VM Density Metrics

- VMs per node (average): N/A
- Highest-density node: N/A
- vCPU:pCPU overcommit ratio: Not computed
- Memory overcommit ratio: Not computed
- Arc-connected VMs:
- VMs using Arc IP fallback:
- Avg CPU utilization (all nodes): 38.3%
- Avg available memory (all nodes): 176 GiB

Storage Pool Capacity

Pool	Raw (GiB)	Usable (GiB)	Used (GiB)	Reserve (GiB)	Safe Alloc (GiB)	Posture
—						
0						
0						
0						
0						
0						
—						

ESU Enrollment

- Eligible Arc-connected VMs:
- Enrolled in ESU:
- Not enrolled in ESU:
- Ineligible:
-

Coverage Assessment

- Monitoring coverage (AMA): 100% (3 of 3 nodes have Azure Monitor Agent)
- Backup coverage estimate: 20% (based on backup items vs VM count)
- Defender for Cloud: Not confirmed or not collected
- WDAC policy: Not collected
- BitLocker status: Not collected
- Certificates expiring in <90 days: 1

Security Posture Summary

- Secured-Core enabled nodes: of 3
- Syslog forwarding nodes:
- RBAC assignments at resource group: 1
- Policy assignments tracked: 2
- Policy exemptions:
- Active Directory site: Not collected

Management Tool Coverage

- WAC installed nodes: of 3
- SCVMM agent nodes:
- SCOM agent nodes:
- Running management services: 4
- Third-party agent types: (0)

Installation Register (Bill of Materials)

Hostname	FQDN	Manufacturer	Model	Serial	BIOS at Deployment	OS Installed	OS Build
azl-iic-n01							
azl-iic-n01.iic.local							
Dell Inc.							
PowerEdge R760							
Not recorded							
2.10.0.0							
Microsoft Azure Stack HCI							
10.0.25398.1189							
azl-iic-n02							
azl-iic-n02.iic.local							
Dell Inc.							
PowerEdge R760							
Not recorded							
2.10.0.0							
Microsoft Azure Stack HCI							

Hostname	FQDN	Manufacturer	Model	Serial	BIOS at Deployment	OS Installed	OS Build
10.0.25398.1189							
azl-iic-n03							
azl-iic-n03.iic.local							
Dell Inc.							
PowerEdge R760							
Not recorded							
2.10.0.0							
Microsoft Azure Stack HCI							
10.0.25398.1189							

Each unit listed above was installed and commissioned as part of this deployment. Serial numbers are recorded as discovered at handoff; missing values indicate the collector could not access the field.

Per-Node Configuration Record

Node	State at Handoff	Logical CPUs	Installed Memory	Domain Joined	BIOS Version
azl-iic-n01					
Up					
64					
512 GiB					
iic.local					
2.10.0.0					
azl-iic-n02					
Up					
64					
512 GiB					
iic.local					
2.10.0.0					
azl-iic-n03					
Up					
64					
512 GiB					
iic.local					
2.10.0.0					

Each node was configured with the values above at the time of deployment.

Network Address Allocation Record

Cluster Network	Role	Network Address	Mask	Metric	State
Management					
Cluster + Client (Management)					
10.0.0.0					
255.255.255.0					
100					
Up					
Storage-1					
Cluster Only (Storage)					
10.0.1.0					
255.255.255.0					
200					
Up					
Storage-2					
Cluster Only (Storage)					
10.0.1.128					
255.255.255.128					
200					
Up					
Workload					
Cluster + Client (Workload)					
10.0.2.0					

Cluster Network	Role	Network Address	Mask	Metric	State
255.255.255.0					
300					
Up					

Cluster networks were assigned and configured as recorded above during deployment.

Storage Configuration Record

Storage Pool	Raw Capacity	Usable Capacity	Health at Handoff
S2D on azlocal-iic-01			
43068 GiB			
—			
Healthy			

Storage pools and their capacities were provisioned at deployment as shown. CSV and virtual-disk details are included in the delivery-registers workbook.

Identity and Security Record

Identity mode	ad
Active Directory site	Not collected
Secured-Core nodes enrolled	of 3
BitLocker	Not collected
WDAC policy	Not collected
Certificates tracked	1
Certificates expiring <90d	1
RBAC assignments at RG scope	1

Azure Integration Record

Tenant ID	00000000-0000-0000-0000-000000000000
Subscription ID	33333333-3333-3333-3333-333333333333
Resource Group	rg-iic-compute-01
Arc-connected machines	1
AKS clusters	1
Azure Monitor Agents	3
Backup items	1
ASR protected items	1

Validation Record

Validation report	See cluster-validation report artifact (Test-Cluster output)
Collectors run	7
Collectors successful	7
Collectors partial	0
Collectors failed	0
Schema validation	Passed
Critical findings	0
Warning findings	2

Known Issues and Deviations

Severity	Item	Deviation	Remediation Path
WARNING			
One or more node certificates expire within 90 days			
Identity posture data indicates certificate on azl-iic-n01 expires within 90 days.			
Review certificate ownership and renew expiring node certificates before handoff.			
WARNING			
iDRAC firmware below recommended baseline on all nodes			
OEM integration data shows iDRAC firmware at 7.10.30.00. The recommended baseline is 7.20 or later.			
Update iDRAC firmware via Dell OME or Lifecycle Controller during next maintenance window.			

Deviations listed below were documented at handoff. Items are accepted as-built unless explicitly marked for follow-up remediation.

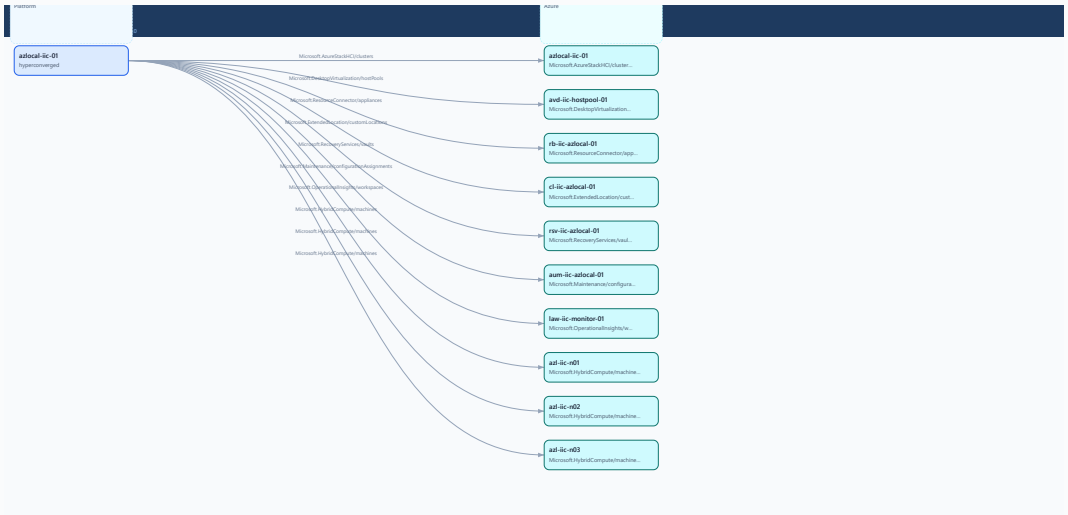
Acceptance and Sign-Off

This as-built package was generated from the Ranger discovery run on the date shown in the Document Control section. Review all findings before accepting this package as the formal handoff record.

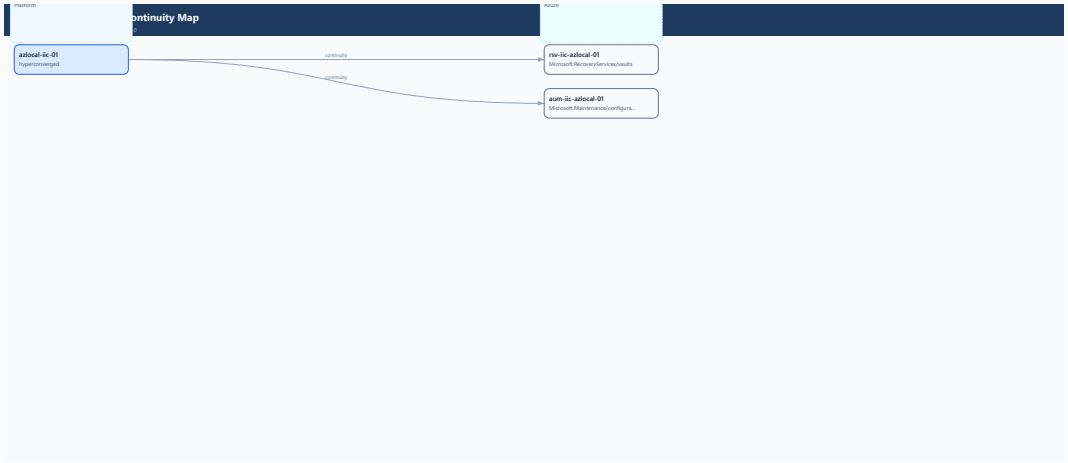
Role	Name	Date	Signature
Implementation Engineer			
Technical Reviewer			
Customer Representative			

Architecture Diagrams

Azure Arc Integration



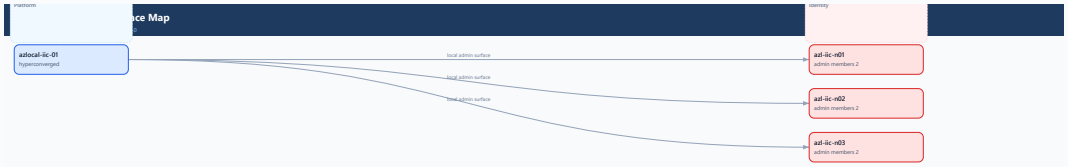
Backup Recovery Map



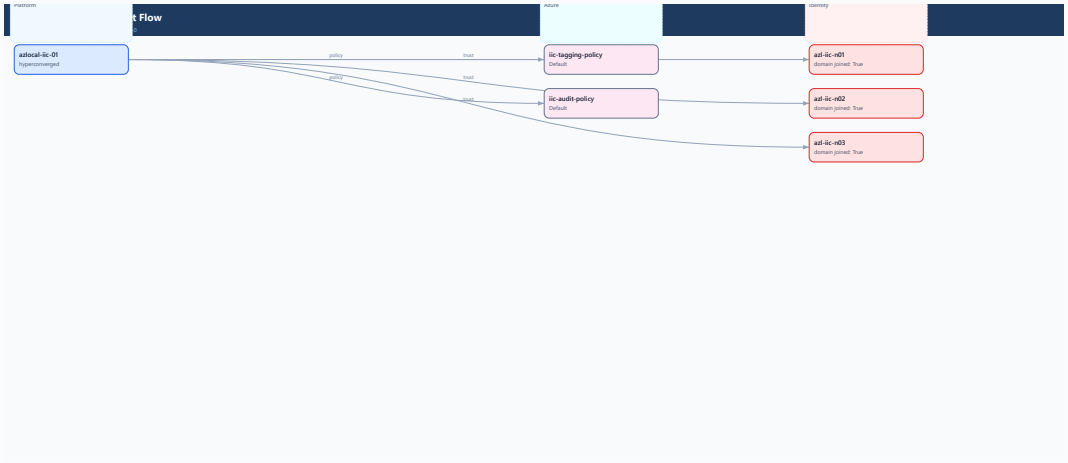
Disconnected Control Plane

Fabric Map

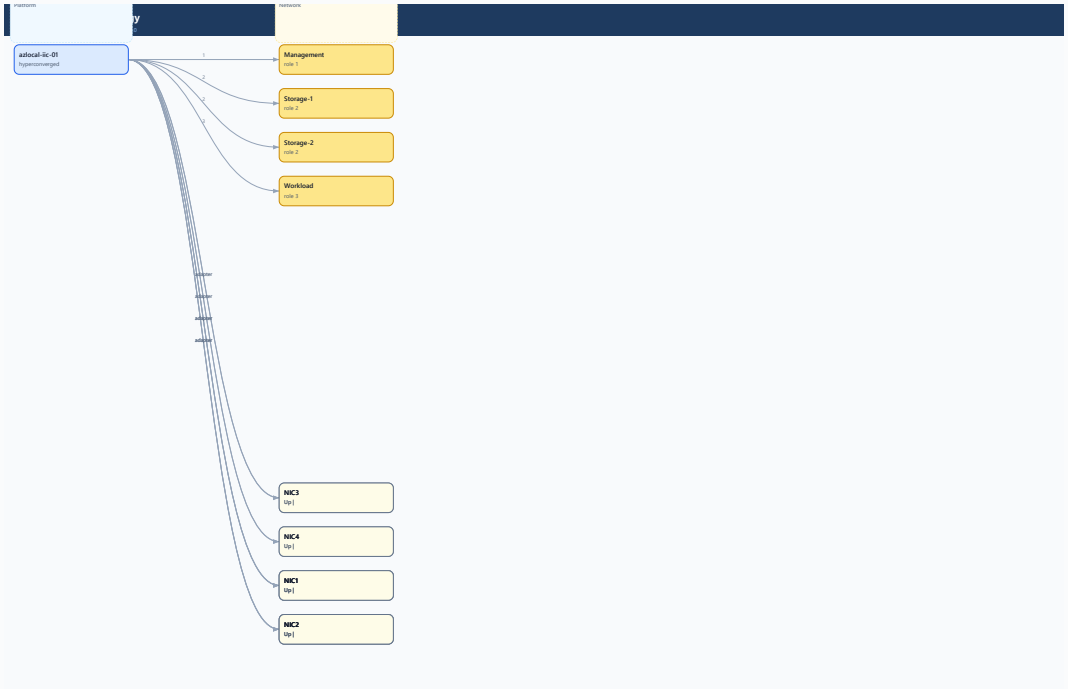
Identity Access Surface



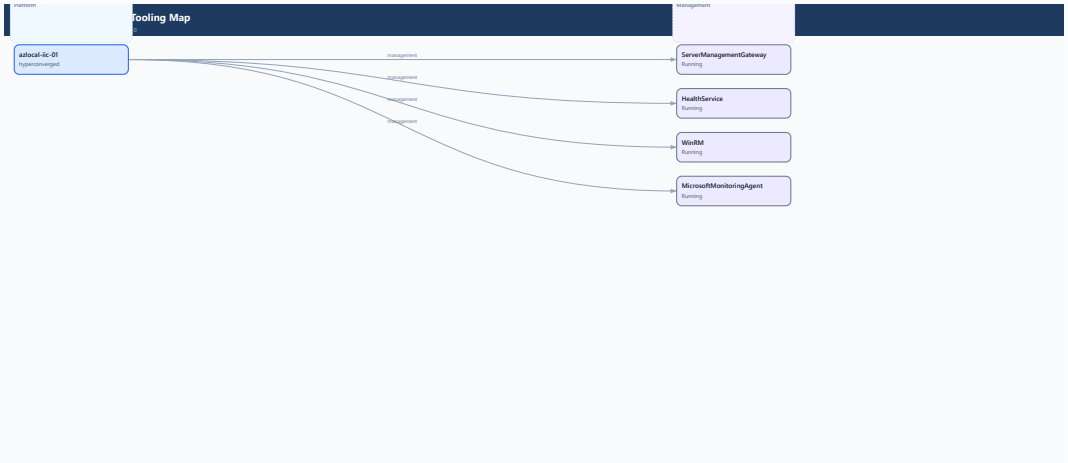
Identity Secret Flow



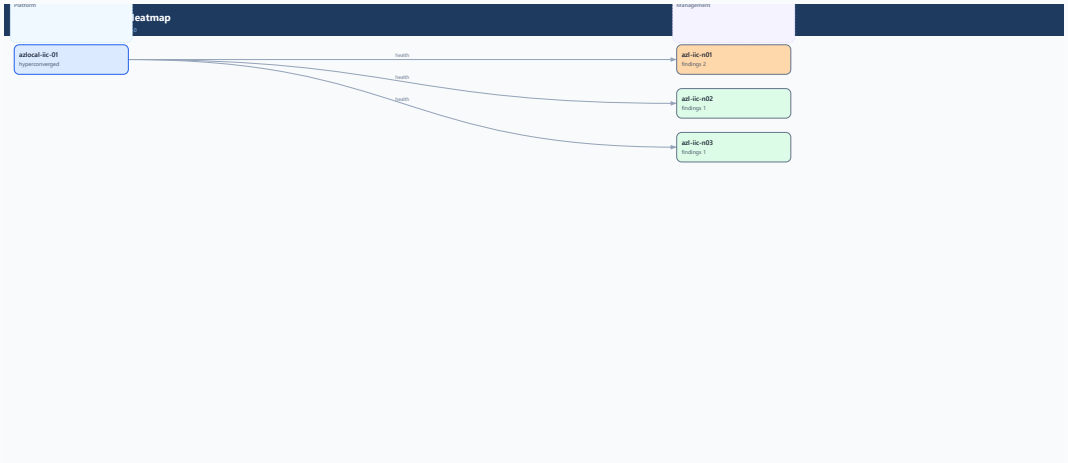
Logical Network Topology



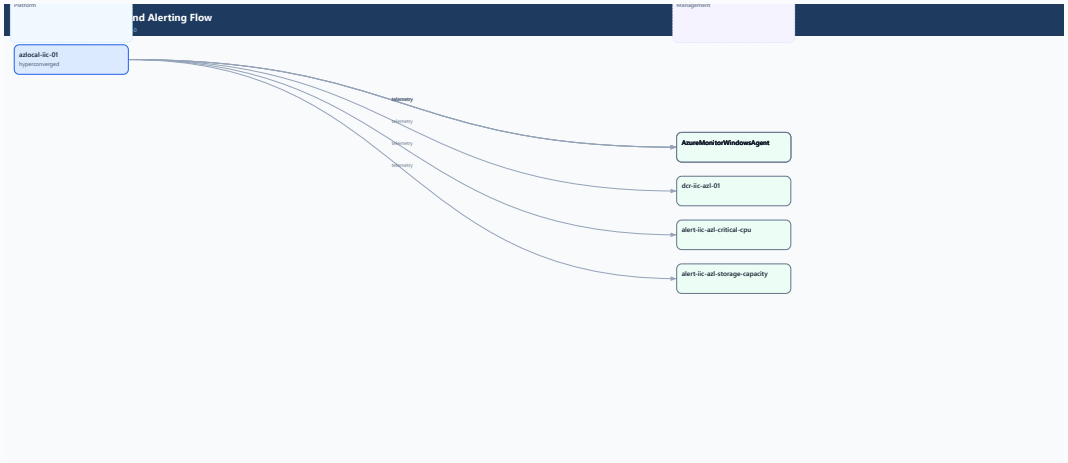
Management Plane Tooling



Monitoring Health Heatmap

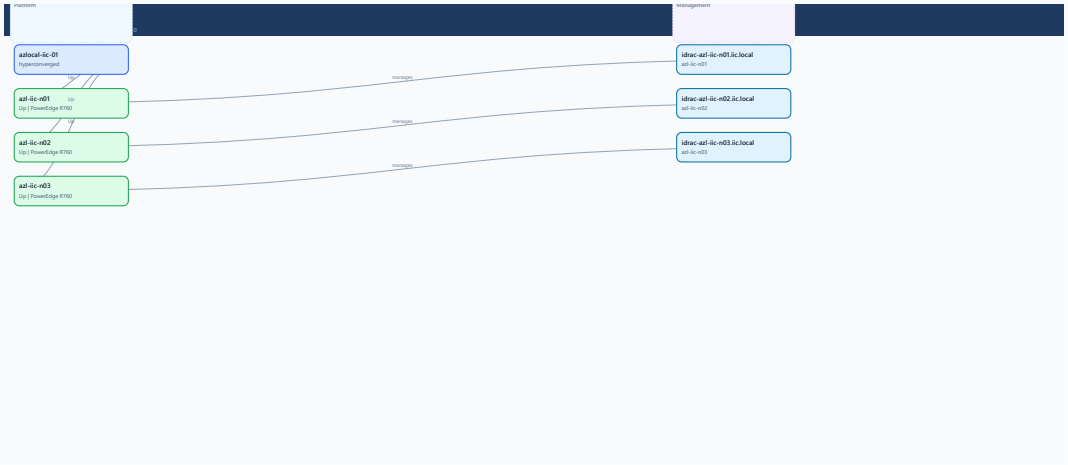


Monitoring Telemetry Flow



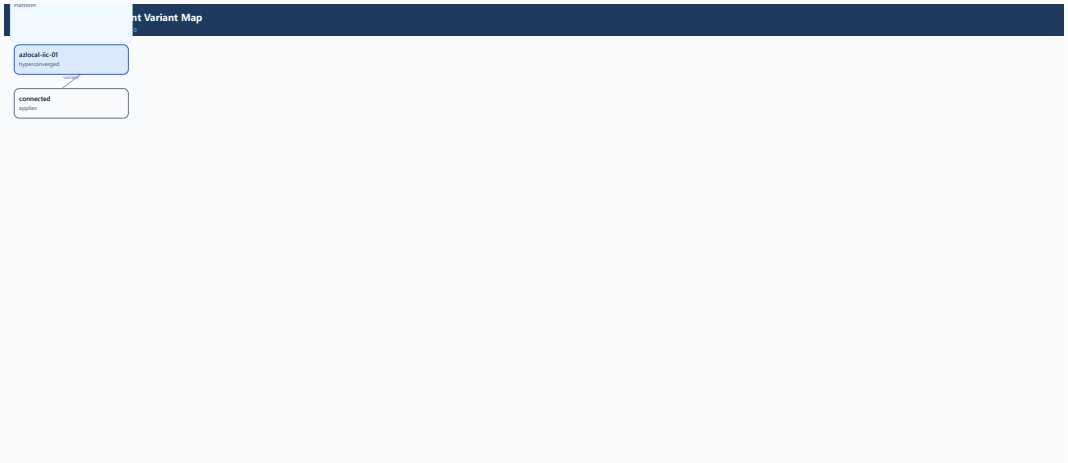
Oem Firmware Posture

Physical Architecture

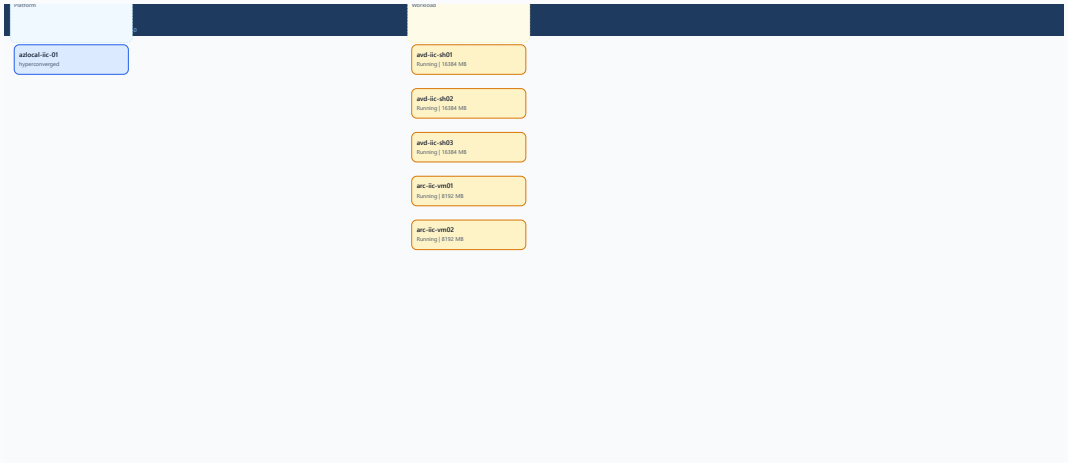


Storage Architecture

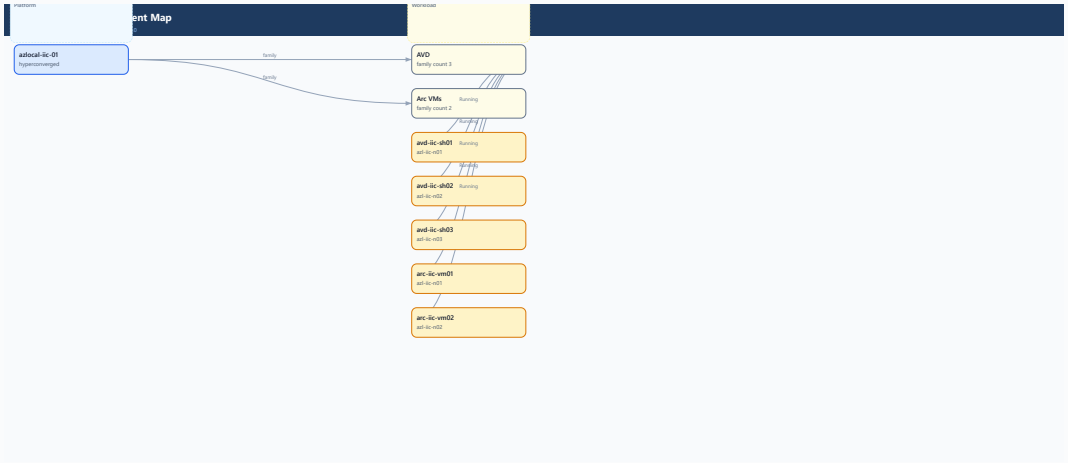
Topology Variant Map



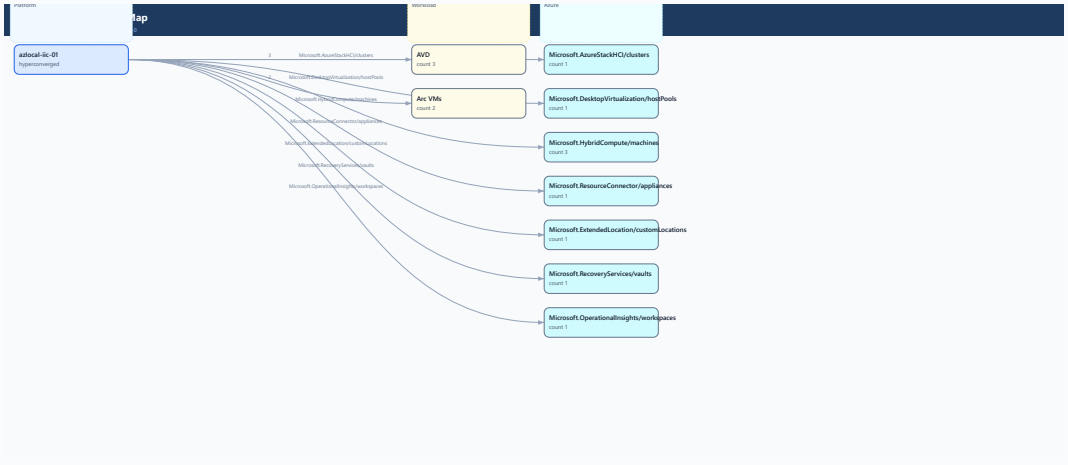
Vm Placement Map



Workload Family Placement



Workload Services Map



Findings

WARNING **One or more node certificates expire within 90 days**

Identity posture data indicates certificate on azl-iic-n01 expires within 90 days.

Current state: 1 certificate expiring within 90 days

Recommendation: Review certificate ownership and renew expiring node certificates before handoff.

WARNING **iDRAC firmware below recommended baseline on all nodes**

OEM integration data shows iDRAC firmware at 7.10.30.00. The recommended baseline is 7.20 or later.

Current state: iDRAC firmware 7.10.30.00

Recommendation: Update iDRAC firmware via Dell OME or Lifecycle Controller during next maintenance window.

INFORMATIONAL **Azure Policy assignments discovered at resource group scope**

2 policy assignments are enforcing tagging and security control auditing.

Current state: 2 policy assignments active

Recommendation: Verify policy scope extends to arc resource group and monitor for compliance drift.

INFORMATIONAL **Cluster event history includes a transient network quorum warning**

A Management network quorum warning event was recorded on 2026-04-06. Cluster recovered automatically.

Current state: single event logged; cluster healthy

Recommendation: Review switch port configuration to confirm management NICs are on dedicated VLANs and bonded correctly.